

Mostafa Gomaa Mahmood

Backend Developer

■ mostafagomaamahmood1999@gmail.com ■ 01013108003 ■ Helwan, Cairo, Egypt
[LinkedIn: mustafa-gomaa-mahmoud](#) | [GitHub: MustafaGomaa1](#)

PROFESSIONAL SUMMARY

A Computer Science graduate with hands-on experience building robust backend systems and RESTful APIs using Java and Spring Boot. Demonstrated ability to design layered architectures, integrate AI/ML microservices, and implement secure authentication flows with JWT. Passionate about clean code, decoupled system design, and delivering scalable solutions.

TECHNICAL SKILLS

Languages	Java, Python, SQL
Frameworks	Spring Boot, Spring Security, Spring Data JPA, FastAPI
Database	MySQL, Hibernate ORM
Auth & Security	JWT (JSON Web Tokens), Role-Based Access Control (RBAC)
Architecture	Layered Architecture, Microservices, REST API Design
Frontend (basic)	Thymeleaf, Bootstrap, HTML, CSS
Tools & Build	Maven, Git, GitHub, Swagger / OpenAPI
AI Integration	ML model serving via FastAPI, cross-service HTTP communication

PROJECTS

Brandos – AI-Powered E-Commerce Platform (Graduation Project)

[Spring Boot](#) · [FastAPI](#) · [MySQL](#) · [JPA](#) · [Thymeleaf](#) · [Python ML](#)

An intelligent e-commerce management platform that combines a full-featured admin backend with a machine-learning recommendation engine in a decoupled architecture.

- Built the Management Core with Spring Boot 3.x handling full CRUD for products, categories, and inventory via a secure admin dashboard.
- Designed and integrated a FastAPI microservice that serves a Python ML model, delivering real-time AI-powered product recommendations on port 5000.
- Implemented cross-service communication: Spring Boot queries the recommendation engine via HTTP, fetches matched product IDs from MySQL, and renders results with Thymeleaf.
- Managed complex relational data mappings using Spring Data JPA and MySQL with hibernate auto-schema updates.

github.com/MustafaGomaa1/brandos-Graduation-Project

Social Media Application

Spring Boot · Spring Security · JWT · MySQL · JPA · Hibernate · Thymeleaf

A full-featured social networking backend with secure authentication, user profiles, posts, and a commenting system, built following clean layered architecture principles.

- Implemented JWT-based authentication and registration flow with Spring Security, protecting all API endpoints with role-based access control.
- Designed RESTful API endpoints (register, login, posts CRUD) documented with Swagger/OpenAPI at `/swagger-ui/index.html`.
- Applied strict Layered Architecture: Controller → Service → Repository → Security layers for clean separation of concerns.
- Persisted data using Hibernate and Spring Data JPA against a MySQL database; server-side views rendered via Thymeleaf and Bootstrap.

github.com/MustafaGomaa1/Social-App

WORK EXPERIENCE

Programming Tutor

iSchool | Jul 2025 – Sep 2025 | Cairo, Egypt

- Taught foundational programming concepts to young students using Code.org's interactive platform.
- Designed age-appropriate lesson plans to introduce computational thinking, loops, conditionals, and event-driven logic.
- Monitored student progress and provided targeted feedback to encourage problem-solving skills.

EDUCATION

Bachelor of Science in Computer Science

Capital University | Graduated 2025